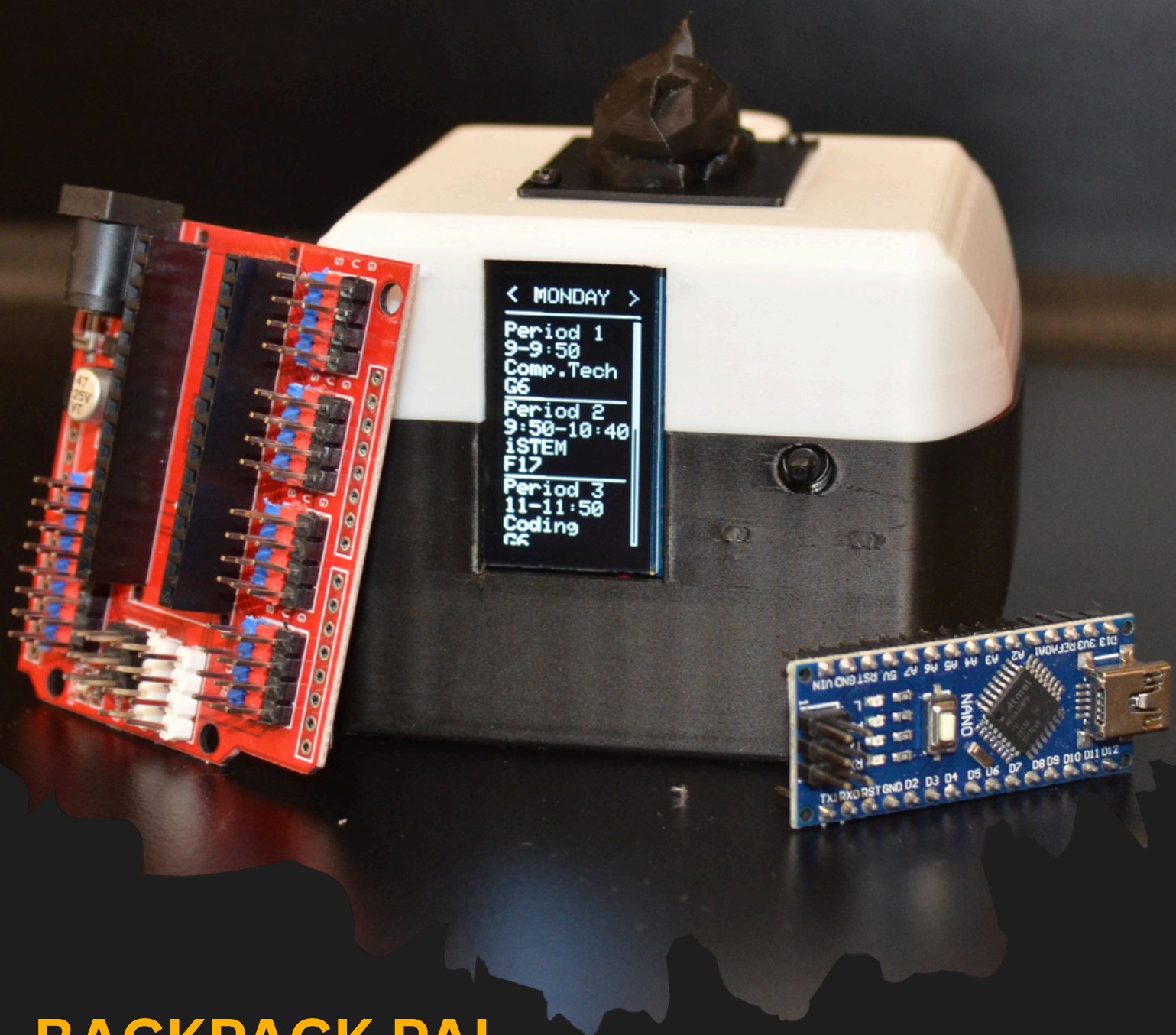




BACKPACK PAL WORKSHOP INFORMATION PACK

Find out more at:
projectbeacon.org.au

ABOUT US

Project Beacon is a not-for-profit organisation based in Sydney whose goal is to bring the world of STEM directly into the hands of young student innovators. All our presenters are current university students pursuing STEM degrees, allowing them to provide an engaging and relatable learning experience for each student. Our interactive and curriculum-aligned workshops aim to teach students the fundamentals of robotics, from manufacturing through to programming and testing. Each workshop is built around students creating robots they can take home, inspiring continual experimentation and future learning.

WHAT IS THE BACKPACK PAL?

The Backpack Pal is a robot designed and 3D printed in-house by our team. It conveniently clips onto the back of a student's backpack and displays their school timetable. The bot has two main functions: a home screen which displays the current time and the upcoming period and a timetable screen wherein students can navigate their weekly timetable with a joystick. There is also a mounting slot on the top for students to customise their bot with a cute animal figurine.

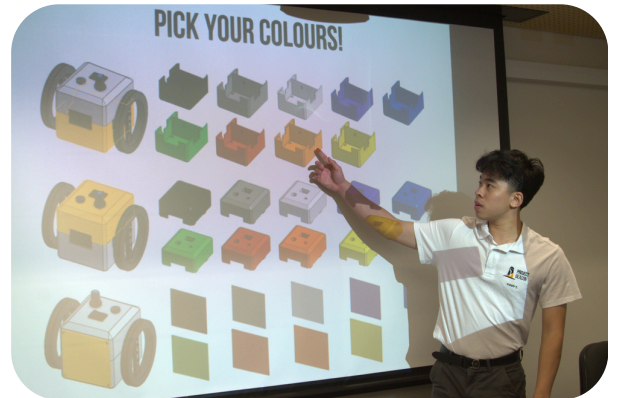


HOW WILL THE WORKSHOP RUN?

The workshop will run for approximately 2 hours and will be scaffolded into the following main sections:

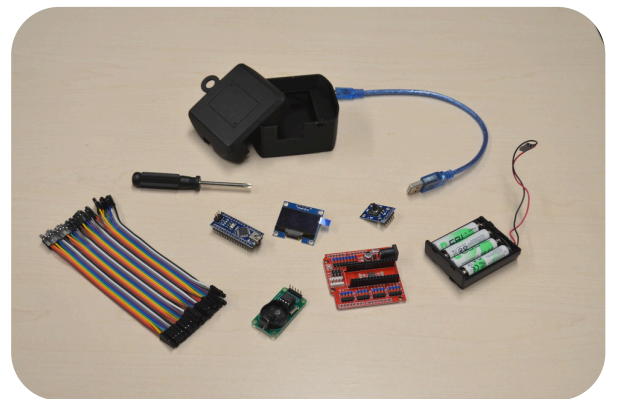
1. Onboarding Presentation (20 Minutes)

Students are introduced to the fundamentals of 3D printing and its unique advantages that made it the ideal manufacturing process for the Backpack Pal's chassis. They are then guided through the inner workings of the bot and how each of its electrical components interact with the Arduino coding language. The presentation concludes with an introduction to the basic principles of circuit building that will be necessary to wire the components together.



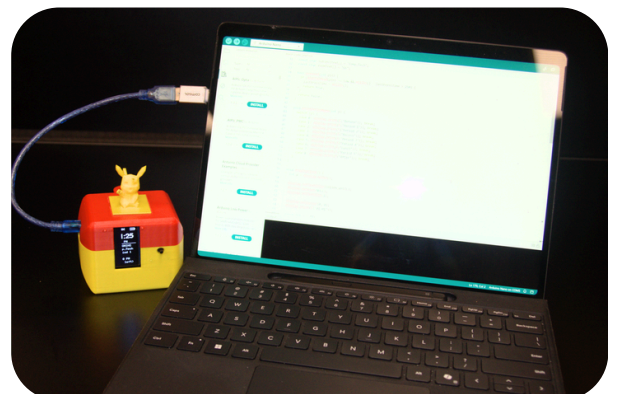
2. Build Session (45 Minutes)

Students begin by customising their Backpack Pal's chassis from a wide variety of colour choices. They then assemble and wire all the components together using our building manual, with support from our team throughout the process.



3. Coding Demonstration (15 Minutes)

A live demo session guides students through the process of setting up Arduino IDE, libraries and any major functions necessary for programming the bot.



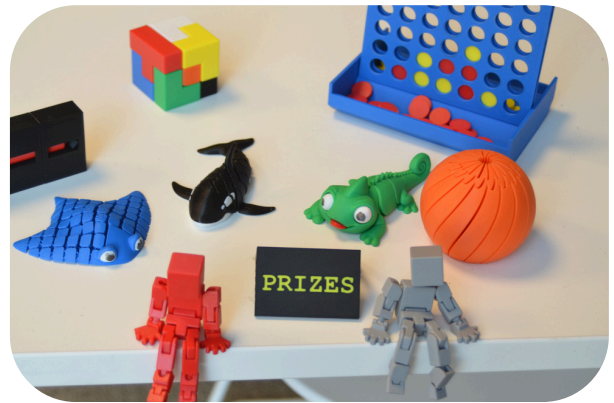
4. Programming and Testing (30 Minutes)

Students apply their knowledge learnt during the coding demonstration to code their Backpack Pal with their own school timetables.



5. Trivia and Q&A Session (15 Minutes)

To conclude the workshop, our team will host an interactive trivia session testing knowledge taught throughout the workshop. Students will have the opportunity to win our 3D-printed prizes. If time is forgiving, we will also be happy to answer any questions in regards to the burgeoning STEM industry and tips to making the most of the university experience.



LEARNING OUTCOMES

All of our workshops are designed with NESAs' curriculum in mind. Though this workshop is specifically designed for the following subjects we are more than happy to host workshops for extracurricular activities (i.e. robotics clubs).

SUBJECT	RELEVANT LEARNING OUTCOMES
Mandatory Technology (Years 7-8)	• TE4-MS-C-01 - explains how materials, systems and components contribute to solutions • TE4-SAF-01 - selects and safely uses tools, materials, technologies and processes • TE4-DIG-02 - uses data and digital systems to code, design and produce projects
iSTEM Elective (Years 9-10)	• ST5-3 - applies engineering design processes to address real-world STEM-based problems • ST5-4 - works independently and collaboratively to produce practical solutions to real-world scenarios • ST5-5 - analyses a range of contexts and applies STEM principles and processes
Computing Technology Elective (Years 9-10)	• CT5-OPL-01 - designs, produces and evaluates algorithms and implements them in a general-purpose and/or object-oriented programming language • CT5-THI-01 - applies computational, design and systems thinking to the development of computing solutions

NEED TO KNOWS

Workshop Details

Price: \$15 per student

Duration: 2 hours

Workshop Type: Incursion

Risk Assessment: [Google Drive Link](#)

Requirements for Schools

Although we provide almost all supplies and a testing arena we do ask that all prospective bookings meet the following requirements:

- There is an available Smartboard/TV screen that can be connected to via a HDMI port or by Microsoft's Screensharing
- All students have readily access to computers/laptops with **Arduino IDE pre-installed**
- Constant supervision by a teacher during the workshop (we will roster enough instructors for an instructor-student ratio of approximately 1 : 5)

Public Liability Insurance

Public Liability Limit: \$20,000,000

Date of Issue: 29/02/2026

Period of Cover: 28/02/2026 to 28/02/2027

Insurer Name: Chubb Insurance Australia Limited

Insurer ABN: 23 001 642 020

AFS Licence No. : 239687

How to Book this Workshop or Contact Us

If you wish to book our Backpack Pal Workshop please click on the "Book Now" button at the top right of our website projectbeacon.org.au and we will get back to you via email within a business day. If you have any further enquiries feel free to email us at support@projectbeacon.org.au. Hope to hear from you soon!